Patrick Thangarajah

MECHANICAL DESIGN | TESTING & VALIDATION | PROTOTYPING | SIMULATION

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Summary

Recent MASc graduate in mechanical engineering with hands-on experience designing, building, and verifying hardware against requirements from concept through validation. Research background in occupant restraint crashworthiness, physical test correlation, DOE, and technical reporting.

Skills

Design & Prototyping	SolidWorks, Onshape, Engineering Drawings, 3D Printing, Lathe, Mill, Bandsaw, Drill Press
Testing & Analysis	DOE, JMP, Minitab, MATLAB, Microsoft Excel, Troubleshooting
Controls & Instrumentation	Arduino, Sensors, Motors, Data Acquisition, Circuit Design, Altium, LTspice
Technical Communication	Scientific literature reviews, technical reports, technical presentations, work instructions, SOPs

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*

- Research in occupant restraint crashworthiness and injury biomechanics
- Graduate coursework average: 90%

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*

- Biomechanics and Medical Devices Option with coursework in human anatomy, physiology, biomaterials, and biomedical design

Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*

- Validated simulations against physical tests across two configurations and 10 response channels, achieving a 0.71 average ISO 18571 correlation score
- Ran a 104-configuration LS-DYNA DOE and used JMP to quantify geometry effects on occupant response, identifying a configuration with 16% lower peak chest deflection
- Troubleshot model-to-test discrepancies in positioning, restraint fit, and response timing to improve agreement with physical tests
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

R&D Engineer (Part-time) — *Remote*
Echoscope — *May 2023 - Sep 2023*

- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functional testing; all five were approved and used by the client

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*

- Developed standardized work instructions and SOPs by reviewing engineering drawings, customer specifications, and existing manufacturing processes
- Reduced fiberglass composite cutting waste by 50% through process analysis and proposed a material-handling workflow projected to reduce pick-and-place time by 50%

Engineering Intern — *Victoria, BC, Canada*
Breathe Global — *Jul 2021 - Aug 2021*

- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components

Projects

Soft Distensible Bladder Model - Capstone Project — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2022 - Apr 2023*

- Prototyped bladder phantoms using silicone and agar, comparing pressure-volume response with literature to guide material selection
- Designed and built the control and data-acquisition interface for a pressurized soft-bladder test system, enabling real-time pressure/volume monitoring and automated filling/voiding cycles
- Verified the system against defined performance requirements, demonstrating stable, repeatable operation up to 4.75 kPa internal pressure

Payload Team Member — *Vancouver, BC, Canada*
UBC Unmanned Aircraft Systems (Student Design Team) — *Sep 2019 - May 2021*

- Designed and prototyped a payload drop system and rover in Onshape and SolidWorks, iterating through 3D printed proof-of-concept builds

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.